

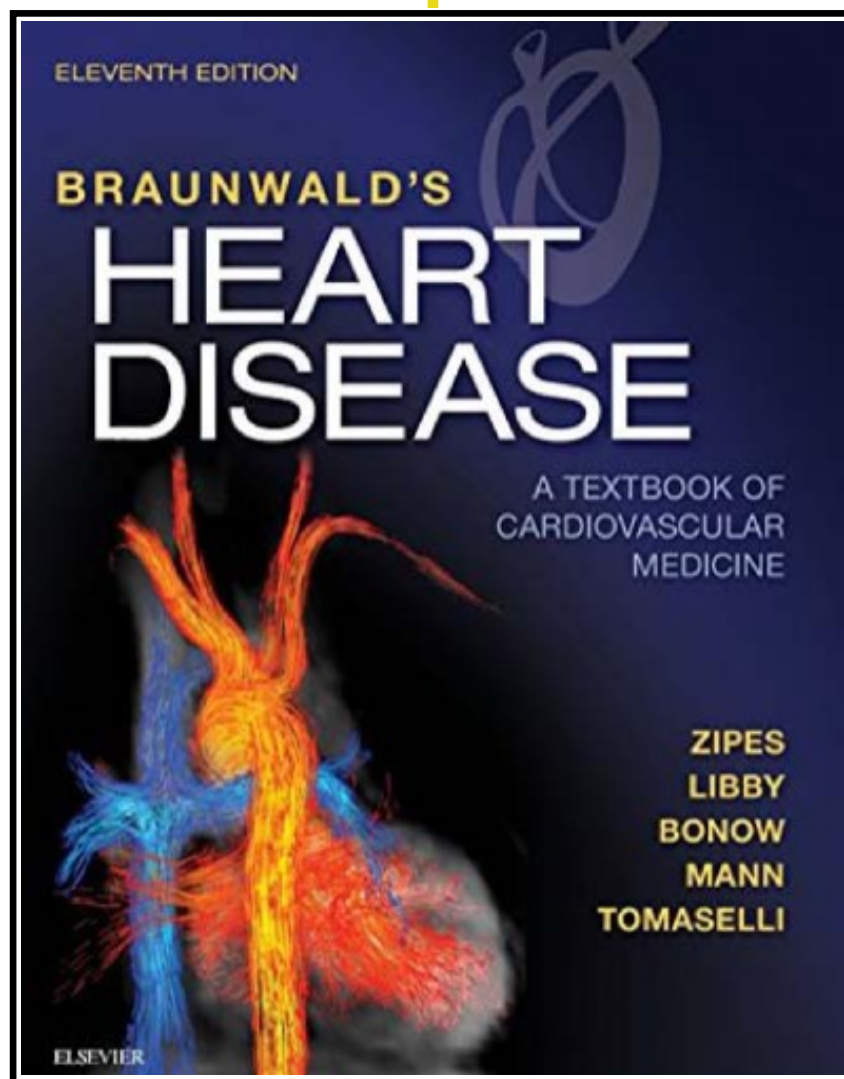
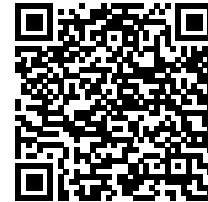
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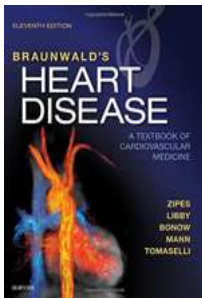
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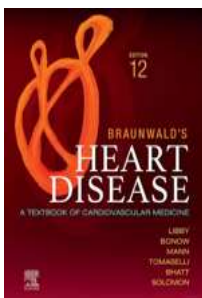
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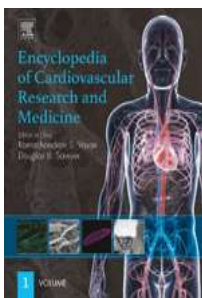
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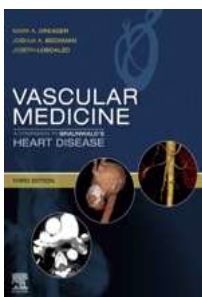
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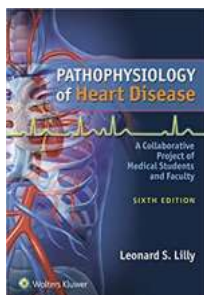
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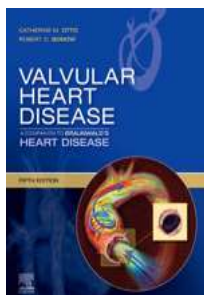
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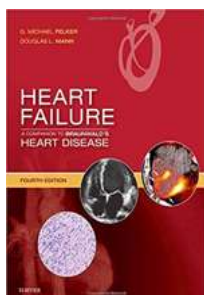
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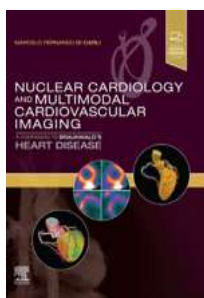
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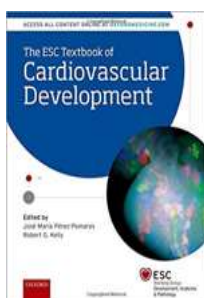
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Ventricular Arrhythmias
Bradyarrhythmias and Atrioventricular Block
Pacemakers and Implantable Cardioverter-Defibrillators
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Preface

This is the 11th edition of the classic textbook, *Heart Disease: A Textbook Of Cardiovascular Medicine*, that Dr. Eugene Braunwald began almost 40 years ago. The editors are pleased and honored to dedicate this edition to him for his extraordinary contributions to the discipline of cardiology, especially this textbook and its companions, and for his unique concept of creating a “living textbook.”

In the past several decades, cardiology has advanced at a breakneck pace on many fronts. Knowledge of the diagnosis and management of patients with heart disease, as well as understanding of responsible mechanisms and preventive approaches, advance daily. Genetics, molecular biology and pharmacology, imaging, catheter-based therapy, and cardiac repair are only a sampling of what we encounter daily.

This constant flow of innovative research has generated a proliferation of new cardiovascular journals, cumulatively publishing an unprecedented amount of information. With the rapidly changing cardiovascular knowledge base, an authoritative textbook like *Heart Disease* to which readers can turn, confident that the statements are as accurate and up-to-date as possible, offers even greater value.

As with each edition of this reference work, international experts — household names to many readers — have totally revised *every* chapter. In addition, 14 new chapters have been added to recognize the ever-expanding role of cardiology into areas such as oncology, chronic lung disease, environmental toxins, catheter-based treatments of congenital heart disease, and other topics. Some sections have been revised for clarity, such as arrhythmias; some have been expanded, such as diseases of heart valves; others have had a shift of emphasis, such as congenital heart disease in the adult. Finally, to maintain vitality of standard topics, new authors have replaced more than a third of those who have tirelessly written for previous editions, in chapters on ethics, personalized and precision medicine, imaging, obesity, diabetes, sleep-disordered breathing, autonomic nervous system, and other areas.

In the preface for the 10th edition of *Heart Disease*, we emphasized that the online version contained audio, video, and written information not available in the print edition. We have continued and expanded this practice. To put this 11th edition in perspective, it contains more than 2700 illustrations and 565 tables, while maintaining the number of printed pages near 2000. The eBook includes an additional 400 illustrations, 60 tables, and 300 videos.

We have divided the book into 11 sections, including fundamentals of cardiovascular disease; genetics and personalized medicine; evaluation of the patient; heart failure; arrhythmias, sudden death, and syncope; preventive cardiology; atherosclerotic cardiovascular disease; diseases of heart valves; diseases of the myocardium, pericardium, and pulmonary vasculature bed; cardiovascular disease in special populations; and cardiovascular disease and disorders of other organs.

We have continued the tradition of including practice guidelines, and have written the text for all levels of learners and for all specialties in cardiology. As before, information not directly relevant to practicing clinicians is presented in smaller font. More detailed information on many topics can be found in the companions to this book:

Cardiovascular Intervention by Deepak L. Bhatt
Cardiovascular Therapeutics by Elliott Antman and Marc Sabatine
Chronic Coronary Artery Disease by James DeLemos and Torbjorn Omland
Clinical Arrhythmology and Electrophysiology by Ziad Issa, John Miller, and Douglas Zipes
Clinical Lipidology by Christie Ballantyne
Diabetes in Cardiovascular Medicine by Darren McGuire and Nikolaus Marx
Heart Failure by Michael Felker and Douglas Mann
Hypertension by George Bakris and Matthew Sorrentino
Mechanical Circulatory Support by Robert Kormos and Leslie Miller
Myocardial Infarction by David Morrow
Preventive Cardiology by Roger Blumenthal, JoAnn Foody, and Nathan Wong
Valvular Heart Disease by Catherine Otto and Robert Bonow
Vascular Medicine by Marc Creager, Joshua Beckman, and Joseph Loscalzo
Braunwald's Heart Disease Review and Assessment by Leonard Lilly
Atlas of Cardiovascular CT by Allen Taylor
Atlas of Cardiovascular MR by Christopher Kramer and W Greg Hundley
Atlas of Nuclear Cardiology by Amil Iskandrian and Ernest Garcia

In keeping with the revitalization theme noted above, one of us (DPZ) will be leaving after this edition. Beginning with the 2nd edition of *Heart Disease* in 1984, Dr. Zipes has written the arrhythmia section and, in more recent editions, with co-authors, and has co-edited *Heart Disease* since the 6th edition. Gordon F. Tomaselli will be his very capable replacement.

The editors and authors, along with the Elsevier staff, have endeavored to make *Heart Disease* the go-to source for current cardiology knowledge, maintaining the high standards Dr. Braunwald set many years ago. We hope readers will enjoy reading this edition and learn from it, as we all strive to improve patient care, our ultimate goal.

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Gordon F. Tomaselli



Preface to the First Edition

Cardiovascular disease is the greatest scourge affecting the industrialized nations. As with previous scourges — bubonic plague, yellow fever, and small pox — cardiovascular disease not only strikes down a significant fraction of the population without warning but also causes prolonged suffering and disability in an even larger number. In the United States alone, despite recent encouraging declines, cardiovascular disease is still responsible for almost 1 million fatalities each year and more than half of all deaths; almost 5 million persons afflicted with cardiovascular disease are hospitalized each year. The cost of these diseases in terms of human suffering and material resources is almost incalculable.

Fortunately, research focusing on the prevention, causes, diagnosis, and treatment of heart disease is moving ahead rapidly. Since the early part of the twentieth century, clinical cardiology has had a particularly strong foundation in the basic sciences of physiology and pharmacology. More recently, the disciplines of molecular biology, genetics, developmental biology, biophysics, biochemistry, experimental pathology and bioengineering have also begun to provide critically important information about cardiac function and malfunction.

In the past 25 years, in particular, we have witnessed an explosive expansion of our understanding of the structure and function of the cardiovascular system—both normal and abnormal—and of our ability to evaluate these parameters in the living patient, sometimes by means of techniques that require penetration of the skin but also with increasing accuracy, by noninvasive methods. Simultaneously, remarkable progress has been made in preventing and treating cardiovascular

disease by medical and surgical means. Indeed, in the United States, a steady reduction in mortality from cardiovascular disease during the past decade suggests that the effective application of this increased knowledge is beginning to prolong human life span, the most valued resource on earth.

To provide a comprehensive, authoritative text in a field that has become as broad and deep as cardiovascular medicine, I enlisted the aid of a number of able colleagues. However, I hoped that my personal involvement in the writing of about half of the book would make it possible to minimize the fragmentation, gaps, inconsistencies, organizational difficulties, and impersonal tone that sometimes plague multiauthored texts. Although *Heart Disease: A Textbook of Cardiovascular Medicine* is primarily a clinical treatise and not a textbook of fundamental cardiovascular science, an effort has been made to explain, in some detail, the scientific bases of cardiovascular diseases.

To the extent that this book proves useful to those who wish to broaden their knowledge of cardiovascular medicine and thereby aids in the care of patients afflicted with heart disease, credit must be given to the many talented and dedicated persons involved in its preparation. I offer my deepest appreciation to my fellow contributors for their professional expertise, knowledge, and devoted scholarship, which has so enriched this book. I am deeply indebted to them for their cooperation and willingness to deal with a demanding editor.

Eugene Braunwald
1980

